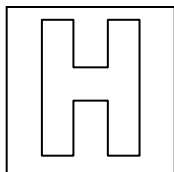


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PIONEER JUNIOR COLLEGE, SINGAPORE

JC2 PRELIMINARY EXAMINATIONS
HIGHER 1

CHEMISTRY

8873/01

Paper 1 Multiple Choice

21 September 2018

1 hour

Additional Materials: Multiple Choice Answer Sheet
 Data Booklet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Write your name, index number and CT group on the Answer Sheet in the spaces provided unless this has been done for you.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

This document consists of **13** printed pages.

Section A

For each question, there are four possible answers labelled **A**, **B**, **C** and **D**. Choose the **one** you consider to be correct.

- 1** In an experiment, 20 cm³ of an organic compound in the gaseous state was sparked with an excess of oxygen. 40 cm³ of carbon dioxide and 20 cm³ of nitrogen were obtained among the products. All gas volumes were measured at the same temperature and pressure.

Which compound would fit the above data?

- A** C₂H₈N₂ **B** C₂H₇N **C** C₆H₅NO₂ **D** C₄H₁₂N₂

- 2** During a laboratory practical, a student tried to determine the concentration of Fe²⁺(aq) and Fe³⁺(aq) in a given mixture through two different experiments.

1. 25.0 cm³ aliquot of the Fe²⁺(aq) and Fe³⁺(aq) mixture solution required 15.00 cm³ of 0.1 mol dm⁻³ of acidified KMnO₄ solution for complete oxidation.
2. Using another 25.0 cm³ aliquot, all the Fe³⁺(aq) was first reduced to Fe²⁺(aq) using zinc metal before requiring 34.20 cm³ of the same KMnO₄ solution for complete oxidation.

What is the concentration of Fe²⁺(aq) and Fe³⁺(aq) in the given mixture?

- A** 0.012 mol dm⁻³ Fe²⁺(aq) and 0.0154 mol dm⁻³ Fe³⁺(aq)
B 0.180 mol dm⁻³ Fe²⁺(aq) and 0.230 mol dm⁻³ Fe³⁺(aq)
C 0.300 mol dm⁻³ Fe²⁺(aq) and 0.384 mol dm⁻³ Fe³⁺(aq)
D 0.300 mol dm⁻³ Fe²⁺(aq) and 0.684 mol dm⁻³ Fe³⁺(aq)

Use of the Data Booklet is relevant to this question.

- 3** Which ion has only one unpaired electron?

- A** Al³⁺ **B** At⁻ **C** Cu⁺ **D** V⁴⁺

- 4 The table shows the fifth ionisation energies of four consecutive elements in the Periodic Table

Element	Q	R	S	T
Fifth ionisation energy / kJ mol^{-1}	32828	37832	9445	10989

What is the formula of chloride of Q?

- A QCl_2 B QCl_3 C QCl_4 D QCl_5

- 5 Which one of the following **cannot** be explained by hydrogen bonds?

- A CH_3CHO has a higher boiling point than C_3H_6 .
 B CH_3CHO is miscible in water.
 C CH_3COOH has a M_r of 120 in benzene.
 D HF has a higher boiling point than HCl.

- 6 Which of the following has the largest bond angle?

- A ClF_3 B H_3O^+
 C SO_2 D NO_3^-

- 7 Which statements concerning the lattice structures of graphite and diamond are correct?

- 1 The C-C-C bond angle between nearest neighbours is smaller in diamond than in graphite.
 2 The shortest carbon-carbon bond occurs in diamond.
 3 All covalent bonds in diamond are of the same strength but those in graphite are not.

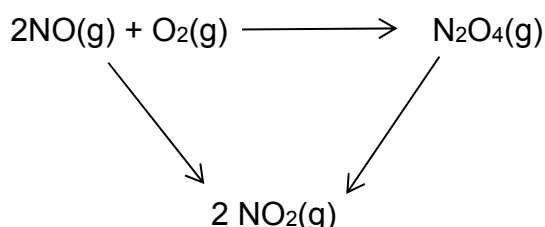
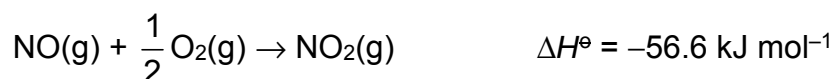
- A 1 only B 2 only C 1 and 3 D 2 and 3

8 Which statements about endothermic reactions are correct?

- 1 There is a net transfer of heat energy from the surroundings to the reacting system.
- 2 On the reaction pathway diagram the products of the reaction are lower than the reactants.
- 3 The total bond energies of the reactants are greater than the total bond energies of the products.

A 1, and 2 B 1 and 3 C 2 and 3 D 1, 2 and 3

9



Given the above data and energy cycle, what is the standard enthalpy change of the following reaction $\text{N}_2\text{O}_4(\text{g}) \rightarrow 2\text{NO}_2(\text{g})$?

A $+57.8 \text{ kJ mol}^{-1}$ B $-57.8 \text{ kJ mol}^{-1}$ C $+114 \text{ kJ mol}^{-1}$ D -114 kJ mol^{-1}

10 The enthalpy change of formation of carbon monoxide and carbon dioxide are given below.

$$\Delta H_f(\text{CO}) = -110 \text{ kJ mol}^{-1}$$

$$\Delta H_f(\text{CO}_2) = -393 \text{ kJ mol}^{-1}$$

Which of the following statements is correct?

- A The enthalpy change of combustion of carbon is -283 kJ mol^{-1} .
- B $\Delta H_f(\text{CO}_2) - \Delta H_c(\text{CO})$ has the same value as enthalpy change of formation of CO.
- C Carbon monoxide is energetically more stable than carbon dioxide.
- D The amount of energy evolved when 1 mole of carbon monoxide is completely burnt is greater than that of carbon.

- 11** An industrial plant illegally dumped some radioactive waste in a landfill.

This waste composed of two radioactive isotopes **X** and **Y** in the proportion 4:1.

The decay of radioactive isotopes follows first-order kinetics.

The half-life of **X** is 2 days whereas that of **Y** is 4 days. By the time the authorities found out about this illegal dumping and analysed a sample of the waste, they found equal amounts of **X** and **Y**.

How long was the waste in the landfill before the authorities arrived?

- A** 2 days **B** 4 days **C** 8 days **D** 16 days
- 12** Using a calorimeter, the following reaction is studied by finding the time taken for a coloured reactant, **F**, to decolourise. The reaction is catalysed by **H**. The following results are obtained.



Expt	Volume of F / cm ³	Volume of G / cm ³	Volume of H / cm ³	Volume of H ₂ O/ cm ³	Time taken/ s
1	10	20	10	10	20
2	10	10	10	20	40
3	10	20	5	15	40
4	5	20	10	15	20

What is the rate equation for the reaction?

- A** rate = $k[\text{G}][\text{H}]$
B rate = $k[\text{F}][\text{H}]$
C rate = $k[\text{F}][\text{G}]$
D rate = $k[\text{F}][\text{G}][\text{H}]$
- 13** Which series is correctly arranged in order of increasing values ?
- A** electrical conductivity of Mg, Al, Si
B electronegativity of P, S, Cl
C melting point of P, S, Cl
D melting point of the chlorides of Na, Mg, Al

- 14 Element **X** is one of the first five elements in Period 3 of the Periodic Table. The following four statements were made about the properties of element **X** or its compounds.

Three statements are correct descriptions and one is false.

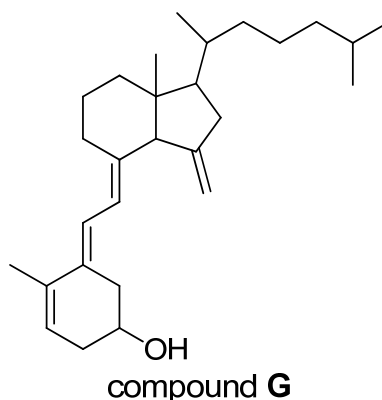
Which statement does **not** fit with the other three?

- A The oxide of **X** dissolves in excess dilute NaOH(aq).
- B The oxide of **X** has a very high melting point.
- C The chloride of **X** reacts with water to give an acidic solution with pH 1.
- D **X** exhibits only one possible oxidation number in its chloride, which is not a solid at room temperature.

- 15 Which of the reagents / tests can be used to differentiate compound 1 and 2?

	Compound 1	Compound 2	Reagents / tests
1	NaCl	MgO	Dilute H ₂ SO ₄
2	SiCl ₄	PCl ₅	damp blue litmus paper
3	AlCl ₃	Al ₂ O ₃	H ₂ O
A	1 only	B 1 and 2	C 2 and 3
			D 3 only

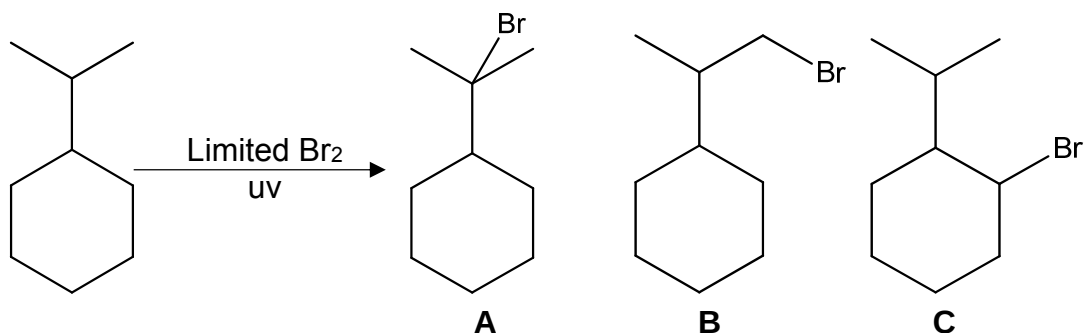
- 16 Compound **G** has the following structure.



How many cis-trans isomers does compound **G** have?

- A 2
- B 2²
- C 2³
- D 2⁴

- 17 Isopropylcyclohexane reacts with bromine to form different mono-substituted products. Three of the products are shown below.



The rate of abstraction of primary, secondary and tertiary hydrogen is given.

Primary	Secondary	Tertiary
$\begin{array}{c} \text{H} \\ \\ \text{H}-\text{C}-\text{H} \\ \\ \text{R} \end{array}$	$\begin{array}{c} \text{H} \\ \\ \text{R}'-\text{C}-\text{H} \\ \\ \text{R} \end{array}$	$\begin{array}{c} \text{R}'' \\ \\ \text{R}'-\text{C}-\text{H} \\ \\ \text{R} \end{array}$
2	3	6

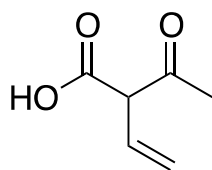
What is the expected ratio of the mono-substituted products **A** : **B** : **C** formed?

- A** 1 : 1 : 1 **B** 1 : 2 : 2 **C** 1 : 6 : 4 **D** 6 : 2 : 3
- 18 Compound **X**, $\text{CH}_2(\text{OH})\text{CH}(\text{CHO})\text{CH}_3$, is heated under reflux with an excess of acidified $\text{K}_2\text{Cr}_2\text{O}_7$ to form compound **Y**. Both **X** and **Y** are separately warmed with Fehling's solution and the observations noted.

What are the observations?

- A** Both **X** and **Y** give a red precipitate.
B Only **X** gives a red precipitate.
C Only **Y** gives a red precipitate.
D Neither **X** nor **Y** gives a red precipitate.

- 19 Which are the products formed when compound **S** reacts with the following reducing agents?

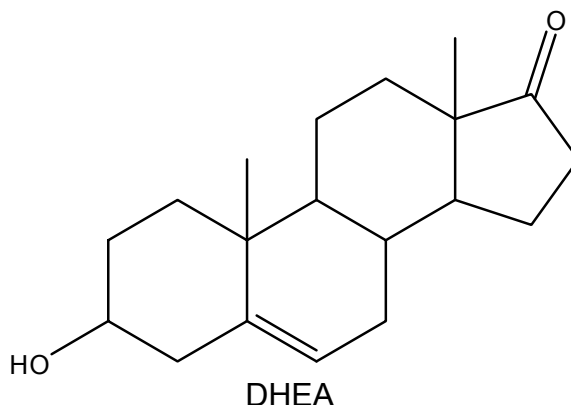


compound **S**

	H₂, Pt	LiAlH₄ in dry ether
A		
B		
C		
D		

- 20 Athletes taking part in the Olympics could be disqualified and have their medal revoked for using banned performance-enhancing drugs.

DHEA, a steroid hormone, is illegally used as a performance-enhancing drug as it has minimal side effects.



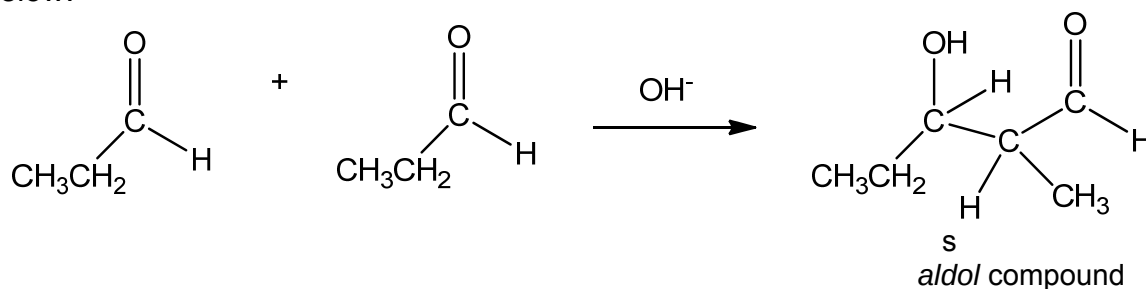
Which reagents would react with DHEA?

- 1 Thionyl chloride, SOCl_2
- 2 Bromine
- 3 Acidified KMnO_4
- 4 Tollens' reagent

A 2 and 4 only B 1 and 3 only C 1, 2 and 3 D 2, 3 and 4

- 21 In the presence of a strong base, aldehydes and ketones may undergo addition reaction to form an *aldol* compound (hydroxycarbonyl).

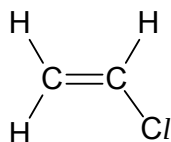
A reaction of propanal with sodium hydroxide to form an *aldo* compound is as shown below.



Which product can be formed when sodium hydroxide is added to a mixture of butanal, $\text{CH}_3\text{CH}_2\text{CH}_2\text{CHO}$, and propanone, CH_3COCH_3 ?

- A $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}(\text{OH})\text{CH}_2\text{CH}_2\text{CH}_2\text{CHO}$
- B $(\text{CH}_3)_2\text{C}(\text{OH})\text{CH}_2\text{CH}_2\text{CH}_2\text{CHO}$
- C $(\text{CH}_3\text{CH}_2)_2\text{C}(\text{OH})\text{CH}_2\text{COCH}_3$
- D $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}(\text{OH})\text{CH}_2\text{COCH}_3$

- 22 The diagram shows the monomer used to make polyvinyl chloride, PVC.

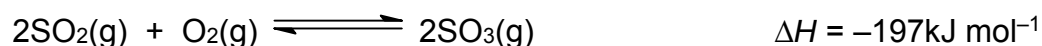


Assuming that one particular molecule of the polymer forms from n molecules of the monomer (where n is many thousands), which statements are correct?

- 1 The relative molecular mass of this polymer molecule is approximately $62.5n$.
- 2 There are n sp^3 carbon atoms in this polymer molecule.
- 3 There are $5n$ σ bonds in one polymer molecule.

A 2 and 3 B 1 and 2 C 2 only D 1 only

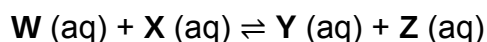
- 23 A reaction involved in the Contact process is shown.



The reaction is investigated at 200 kPa and 700K and the value of the equilibrium constant, K_c , is found to be Y. The reaction is then investigated at 1000 kPa and 700K and the value of K_c is found to be Z. Which statement comparing Y and Z is correct?

- A Y and Z are the same.
- B Y is greater than Z.
- C Z is 2.2 times greater than Y.
- D Z is 5.0 times greater than Y.

- 24 An equilibrium can be represented by the equation below.



In 0.50 dm^3 of a certain mixture, the concentrations of these substances at equilibrium were as shown.

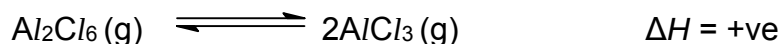
Species	W	X	Y	Z
Concentration / mol dm^{-3}	1.00	2.00	2.00	1.00

A certain amount of Y was added to this equilibrium mixture.

What is the number of moles of Y added such that the new equilibrium [W] is 1.20 mol dm^{-3} ? Assume that the temperature remained constant throughout.

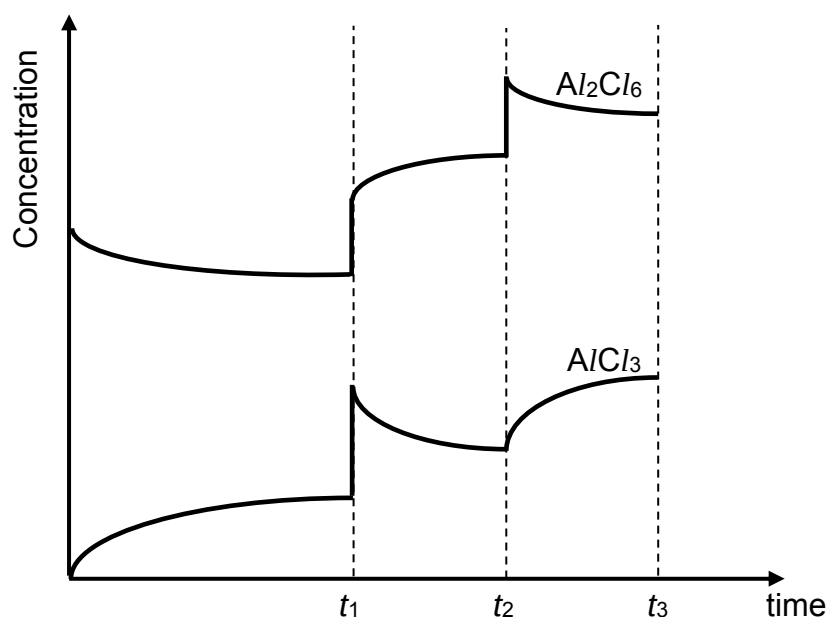
- A 0.10 B 0.20 C 0.75 D 1.50

- 25 Aluminium chloride exists in two different forms in the vapour state. When some Al_2Cl_6 was added into a reaction vessel, the following equilibrium is slowly set up.



At different times during the experiment, changes were made to the conditions in the reaction vessel. At each time, there was only one change made to the conditions in the reaction vessel.

The change in the concentrations in the mixture with time is shown in the graph below.



Which conclusion can be drawn from the graph?

- A At t_2 , a catalyst was added to the system.
- B The temperature of the mixture at t_2 was higher than that at t_1 min.
- C The value equilibrium constant, K_c , is different at t_1 and t_3 .
- D At t_1 , the pressure of the system was increased by decreasing the volume.

- 26 Values for the ionic product of water, K_w , at two different temperatures are given below

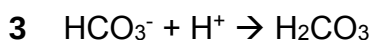
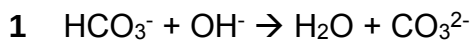
Temperature / °C	K_w / mol ² dm ⁻⁶
25	1.00×10^{-14}
62	1.44×10^{-14}

Which statement is correct for pure water at 30°C?

- A pH < 7
 B pH = 7
 C $[H^+] > [OH^-]$
 D $[H^+] = 1.44 \times 10^{-7}$
- 27 Which aqueous mixture will resist changes in pH on the addition of small amounts of an acid or a base?
- A ammonia + sodium hydroxide
 B hydrochloric acid + propanoic acid
 C propanoic acid + sodium propanoate
 D sodium propanoate + sodium hydroxide
- 28 A mixture is made by adding 20 cm³ of 0.1 mol dm⁻³ of H₂SO₄ with 80 cm³ of 0.1 mol dm⁻³ NaOH. What is the pH of the final solution at 25°C?
- A 11.60
 B 11.80
 C 12.60
 D 12.80

- 29** In humans it is important for blood to be maintained at a pH between 7.35 and 7.45. One of the ways it does this is by using a buffer of $\text{CO}_2(\text{aq})$ and $\text{HCO}_3^-(\text{aq})$.

What are the equation(s) that represent how the blood acts as a buffer when small amounts of acid and alkalis are added?



A 1 only

B 2 only

C 1 and 3

D 2 and 3

- 30** The gecko, a small lizard, can climb up a smooth glass window. The gecko has millions of microscopic hairs on its toes and each hair has thousands of pads at its tip. The result is that the molecules in the pads are extremely close to the glass surface on which the gecko is climbing. What is the interaction between the gecko's toe pads and the glass surface?

A dative bonds

B effective nuclear charge

C instantaneous dipole–induced dipole attractions

D permanent dipole dipole attractions

Answers

1	A	11	C	21	D
2	C	12	D	22	D
3	D	13	B	23	A
4	B	14	A	24	C
5	A	15	D	25	D
6	D	16	B	26	A
7	A	17	B	27	C
8	B	18	B	28	C
9	A	19	C	29	D
10	B	20	C	30	C

7 – A

7 – B

8 – C

8 – D